MILITARY SPECIFICATION

DYE, DIETHYLAMINO-AZOBENZENE (FOR COLORED SMOKE MIXTURES)

1. SCOPE

1.1 This specification covers one type of dye, diethylamino-azobenzene for use in colored smoke mixtures.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids, or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

PPP-D-723 - Drums, Fiber

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes (ABC-STD-105)
MIL-STD-109 - Quality Assurance Terms and Definitions
MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-1168 - Lot Numbering of Ammunition
MIL-STD-1235 - Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes

(Copies of specifications, standards, drawings and publications required by contractor in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

FSC: 6810

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AMERICAN SOCIETY FOR TESTING AND MATERIALS

ASTM-E324 Relative Initial and Final Melting Points and the Metling Range of the Industrial Chemicals
ASTM-E300 Sampling Industrial Chemicals

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Penn., 19103).

3. REQUIREMENTS

3.1 Diethylamino Azobenzene. The diethylamino-azobenzene dye shall be in accordance with Table I.

TABLE I

Physical Properties	Requirement	Test Method
Volatile matter, percent Maximum, (max.)	0.2	4.4.1
Purity,percent Minimum (min)	95.0	4.4.2
Sodium Chloride percent,(max)	0.3	4.4.3
Melting point degrees centigrade,°C (min)	94.5	4.4.4

3.2 Workmanship. The dye shall be free of dirt, chips, grease and other foreign matter.

3.3 First article testing. This specification makes provisions for first article testing. Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

4. QUALITY ASSURANCE PROVISIONS

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4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order,the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspection set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to MIL-STD-109 to define terms used herein.

4.1.1 Submission of product. At the time each completed lot of items deliverable under the contract is submitted to the Government for acceptance, the contractor shall supply the following information accompanied by a certificate which attest that the information provided is correct and applicable to the product being submitted:

a. A statement that the lot complies with all of the quality assurance provisions specified in this specification.

b. Specification number and date, together with identification and date of changes thereto.

c. Certificates of analysis on all materials used directly by the contractor when such materials is controlled by Government specifications shall be made available upon request by the Contracting Officer.

d. Quantity of product in the lot.

e. Date submitted.

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.2 First article inspection

4.2.1 Submission. Prior to the start of regular production the contractor shall submit a first article sample (see 6.1) to a Government approved facility as

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designated by the contracting officer for evaluation in accordance with the provisions of 4.2.2. The first article shall consist of 1 pound of dye in accordance with instructions issued by the Contracting Officer. All samples submitted shall have been produced by the contractor using the same production processes, procedures, and equipment as will be used in fulfilling the contract. All materials, including packaging and packing, shall be obtained from the same sources of supply as will be used in regular production. The sample shall be accompanied by certificates of analysis. A first article quantity, or portion thereof, as directed by the Contracting Officer, shall also be submitted whenever there is a lapse in production for a period in excess of 90 days, or whenever a change occurs in manufacturing process, material used, drawing, specification or source of supply as to significantly affect production uniformity as determined by the Government. Prior to submission, the contractor shall inspect the sample to the degree necessary to assure that it conforms to the requirements of the contract and submit a record of this inspection with the sample. A sample containing known defects will not be submitted unless specifically authorized by the Contracting Officer.

4.2.2 Inspections to be performed. The sample will be subjected by the Government to any or all of the examination or tests specified in 4.4 of this specification.

4.2.3 Rejection. If any sample fails to comply with any of the applicable requirements, the first article quantity shall be rejected. The Government reserves the right to terminate its inspection upon any failure of a sample to comply with any of the stated requirements.

4.3 Inspection provisions

4.3.1 Lot formation. A lot shall consist of one or more batches, of dye produced by one manufacturer, in accordance with the same specification, or same specification revision, under one continuous set of operating conditions. Each lot shall consist of that quantity of dye that has been subjected to the same unit chemical or physical mixing process intended

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to make the final product homogeneous. The product shall be submitted for inspection in accordance with MIL-STD-105. The criteria and procedures for the assignment of lot numbers shall be in accordance with Standard MIL-STD-1168.

4.3.2 Examination. Sampling plans and procedures for the following classifications of defects shall be in accordance with MIL-STD-105 (ABC-STD-105), except that inspection. For critical defects shall be 100 percent. Contractor's sampling plans, if used, shall be approved by the Government and shall provide, as a minimum, the protection afforded the Government by the sampling plans in MIL-STD-105. Continuous sampling plans in accordance with MIL-STD-1235 may be used if approved by the procuring activity. Also, at the option of the procuring activity, AQL's and sampling plans may be applied to the individual characteristics listed, using an AQL of 0.40 percent for each Major defect, and an AQL 0.65 percent for each Minor defect, except where 100 percent inspection is specified.

4.3.2.1 Drums, prior to filling (Polyethylene liner)

Categories	Defects	Method of Inspection	Code No. (See 6.2)
Critical: None defined.			
Major:	AQL 0.40 percent		
101.	Liner cut, torn or punctured....	Visual	01001
Minor: None defined			

4.3.2.2 Drums, prior to closing (Polyethylene liner)

Categories	Defects	Method of Inspection	Code No.
Critical: None defined			
Major:	AQL 0.40 percent		
101.	Liner not completely sealed.....	Visual	02001

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Minor: None defined

4.3.2.3 Drum, sealed (fiber)

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 0.40 percent		
101.	Weight of contents	Scale	03001
102.	Closure of incomplete or damaged to the extent that contents sift out	Visual	03002
Minor:	AQL 0.65 percent		
201.	Markings misleading or unidentifiable	Visual	03003

4.3.3 Sampling. Samples from individual containers shall be selected in accordance with section 23 to 31 of ASTM E300, using a sampling thief. Samples shall be identified to provide identification of the containers used to provide the sample. For the Tests given in 4.4.1 thru 4.4.3 the containers in the lot shall be divided into two (2), approximately, equal groups. For each of these 2 groups, an individual sample shall be prepared in the following manner: Randomly select containers for sampling in accordance with MIL-STD-105. However, if each group contains less than five(5) containers, select all containers. Approximately equal portions of the dye shall be taken from each selected container, in such quantity that the aggregate weight of the portions of the samples from each group shall be approximately one (1) pound. Combine these portions and thoroughly mix. Use the two samples formed to perform the tests described in 4.4.1 thru 4.4.3. For the test given in 4.4.4 the containers in the lot shall be divided into four(4), approximately, equal groups. For each group, a sample shall be prepared in the same manner as described above. If any of the above samples fail to meet any of the requirements specified in Section 3, the lot shall be rejected. The classification and code numbers for the tests shall be as follows:

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Tests	Defects	Code No.
Volatile Matter	Major	04001
Purity	Major	05001
Sodium Chloride	Major	06001
Melting Point	Major	07001
Workmanship	Minor	08001

4.4 Test procedures

4.4.1 Determination of volatile matter. An accurately weighed portion of approximately 5 grams (gm) of the dye shall be transferred to a tared, glass-stoppered weighing bottle, 50 to 75 millimeters (mm) in diameter by 25 to 30 mm in height. The bottle and contents shall be weighed accurately, dried at 60°C ± 2°C overnight cooled to room temperature in a desiccator and reweighed. The percentage of volatile matter shall be calculated from the loss in weight. The dried sample shall be retained for the following determinations.

4.4.2 Determination of purity. An accurately weighed portion of approximately 0.50 gm of the dye previously dried overnight at 60° ± 2°C. shall be transferred to a 600 milliliter (ml) beaker and about 200 ml of 95 percent ethyl alcohol shall be added. The contents of the beaker shall be heated on a steam bath until as much of the sample as possible has dissolved. The supernatant liquid shall be decanted into a 500 ml volumetric flask. An additional 50 ml of alcohol shall be added to the residue contained in the beaker. The residue shall be heated and decanted as before. This procedure shall be repeated until the supernatant liquid is colorless. The contents of the flask shall be diluted to the mark with 95 percent, alcohol, the temperature of the solution adjusted to room temperature and again diluted to the mark (solution "A"). The contents of the flask shall be mixed throughly. Using a pipette, a 10 ml aliquot of the solution "A" shall be transferred to a 200 ml volumetric flask and diluted to the mark with alcohol as before, (solution "B"). In the same manner, a 10 ml aliquot of solution "B" shall be diluted to 100 ml with alcohol (solution "C"). The absorbency of solution "C" shall be determined at wavelengths of 336

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millimicrons (mu) 414 mu, and 600 mu using a blank of 95 percent ethyl alcohol taken from the same lot that was used in the preparation of the sample solutions. A wavelength calibration should be made against suitable spectral lines, using a mercury source at 400 mu region, and a hydrogen source at the alpha hydrogen line of 650.3 mu (see 6.4). The diethylaminoazobenzene content shall be calculated as follows:

$$\text{Percent diethylamino-azobenzene} = \frac{A336}{122.5)b)\ (c)} \times 100 \text{ (or) } \frac{A\ 414 \times 100}{122.5(b)\ (c)}$$

$$\text{or } \frac{A600 \times 100}{122.5(b)\ (c)}$$

Where:

A336 = absorbency of solution at 336 mu.
A414 = absorbency of solution at 414 mu.
A600 = absorbency of solution at 600 mu.
b = cell depth, centimeters (cm)
c = concentration of sample, gm/liter = 0.01 times weight of sample
122.5= aborbancy index of diethylaminoazobenzene

4.4.3 Sodium chloride content. An accurately weighed portion of approximately 0.3 gm of the dye previously dried for 2 hours at 60 ± 2°C., shall be transferred to a 400 ml beaker and moistened with a few ml of 95 percent ethyl alcohol. Approximately 250 ml of distilled water shall be added and stirred for a few minutes to dissolve the sodium chloride. The suspension shall be filtered, using a No. 40 Whatman filter paper or equivalent. The residue on the filtered paper shall be washed with five 10 ml portions of distilled water directed from a wash bottle, adding the washings to the filtrate. Five ml of concentrated nitric acid and 15 ml of 0.1N silver nitrate ($AgNO_3$) solution shall be added to the filtrate. The suspension shall be heated almost to boiling, then allowed to stand in the dark for 4 hours. The supernatant liquid shall be tested for completeness of precipitation with a few drops of $AgNO_3$ solution, then the precipitate

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collected on a tared medium porosity sintered glass crucible. The precipitate on the crucible shall be washed with 0.01N nitric acid solution until the washings show no turbidity when tested with one or two drops of 0.1N hydrochloric acid solution. The precipitate shall be washed with several 10 ml portions of acetone to dissolve out any occluded dye. The crucible shall be aspirated until the odor of acetone can no longer be detected. The crucible and contents shall be dried for 2 hours at 135° to 140°C., cooled in a desiccator and weighed. The sodium chloride content shall be calculated as follows:

$$\text{Percent sodium chloride} = \frac{A \times 40.75}{W}$$

Where:
A = gain in weight of crucible
W = weight of sample

4.4.4 Melting point. The melting point shall be determined in accordance with ASTM Designation E324.

5. PREPARATION FOR DELIVERY

5.1 Packing

5.1.1 Drums. Unless otherwise specified in the contract or purchase order, the dye shall be packed in non-returnable fiber durms in accordance with Specification PPP-D-723, (Type III, Grade A). The drums shall be finished with a nominal 0.004 inch thick polyethylene liner, properly heat sealed or otherwise closed to afford protection to the contents equivalent to that given by a heat seal closure.

5.1.2 Level C. The dye shall be packed to afford protection against damage during direct shipment from supply source to the first receiving activity or immediate use. Containers shall comply with Uniform Freight Classification Rules and Container Specifications for rail shipments or National Motor Freight Rules and Container Specifications for truck shipment as applicable.

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5.2 Marking. In addition to any special markings required by the contract or purchase order, containers shall be marked in accordance with MIL-STD-129. Marking shall include, but is not limited to the following information:

a. Manufacturers name
b. Product designation
c. Lot number
d. Date of manufacture
e. Number of this specification

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

a. Title, number, class and date of this document.
b. Provision for submission of first article samples

6.2 Inspection code numbers. The five digit code numbers assigned to the inspection herein are to facilitate future data collection and analysis by the Government.

6.3 Diethylamino-azobenzene is listed as Keco Oil Yellow 2635 by Keystone Amiline and Chemical Co. 321 N. Loomis St. Chicago, Illinois.

6.4 A calibrated Beckman quartz spectrophotometer was found to be satisfactory for this measurement. Instructions for the calibration of the Beckman Model DU Bulletin 89-E, "Instructions for the Ultraviolet Accessory Set, "Beckman Instruments, Inc.

CUSTODIAN:
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NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

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   A. GIVE PARAGRAPH NUMBER AND WORDING.

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